

Mandibular Molars

- They are 3 in each quadrant
- Each molar has 2 roots (Mesial root – Distal root), the mesial root is the longer.
- Roots are widely divergent ?? to give maximum anchorage against forces.
- Each molar has five surfaces.
- width > thickness
- All Mandibular molars have lingual convergence and lingual inclination.
- All mandibular molars have distal convergence .

1 (7y) **2** (8y) **3** (10-12y) **4** (10-11y) **5** (10-12y) **6** (6y) **7** (12-13y) **8** (18y)

1 (6y) **2** (7y) **3** (9-10y) **4**(10-12y) **5** (11-12y) **6** (6y) **7** (11-12y) **8** (18y)

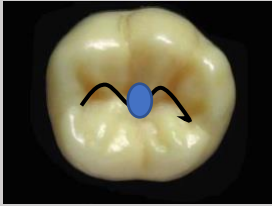
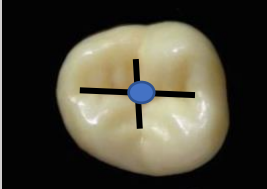
	Beginning of calc.	Crown completed	Eruption	Root completed
<u>6</u> 6	At birth At birth	-3 years	6-7 y 6-7 y.	
<u>7</u> 7	2½-3 y. 2½-3 y.		12 y. 12 y.	
<u>8</u> 8	7-9 y. 8-10 y.		18 y. 18 y.	

Buccal aspect	Lower 6		Lower 7		Lower 8	
The geometric outline	Trapezoidal with small uneven side present cervically.					
Mesial outline :	concave till the contact area at the junction of the occlusal and middle 1/3.					
Distal outline:	straight till the contact area.(more cervically)				No DCA	
Occlusal outline:	formed by 3 cusps, MB, DB, D (sometimes 2 L cusps could be seen behind the B cusps)		MB cusp is longer than the DB cusp.		*The 3 rd molar is either 4 or 5 cusp type (MBC is the widest & highest of the B cusps)	
Cervical outline:	straight with dipping into the root bifurcation					
Anatomical landmarks: A- Elevations	* Cervical ridge at the cervical 1/3					
B- Depressions	* 2 Developmental grooves (MB, DB) * Developmental depression run M-D in the middle 1/3		Buccal developmental groove may end in a buccal pit.			
<u>The Roots:</u>	*The roots twice the length of the crown. * The axis of the roots are inclined distally * Root trunk: 3 mm. * Deep developmental groove extend on the root trunk.		The roots are closer and nearly parallel with more pointed apex and more distal inclination than 6		* 2 Short roots (M and D) may be fused in all or part of their length -*The roots are more inclined distally than 1 st and 2 nd molars	
<u>Mesial root</u>	Cervical half curved mesially & its apical half curved distally . -Apex is located below MBC. It is longer than DR by 1 mm.					
<u>Distal root</u>	-Straighter than the MR -Apical third is curved distally					

LINGUAL ASPECT	Lower 6	Lower 7	Lower 8
			
Mesial outline:	Straight till the contact area at the junction of the occlusal and middle 1/3		
Distal outline:	Convex		No DCA
Occlusal outline:	formed by 2 lingual cusps, ML, DL of nearly equal width(part of the D cusp could be seen)	The lingual cusps are more pointed than buccal cusps.	
Cervical outline:	Irregular with dipping into the root bifurcation		
Anatomical landmarks:	Lingual Developmental groove		
* convergence	Lingual convergence		
The roots	The root trunk appears longer & narrower due to: * The occlusal positioning of the cervical line	Taller root trunk with a developmental depression.	
convergence	* Lingual convergence		

Mesial Aspect	Lower 6	Lower 7	Lower 8
			
The geometric outline	Rhomboid due to lingual inclination .		
Buccal outline:	convex cervically represents Cervical Ridge, then slightly concave represents middle depression buccally finally ends by convexity related to the cusps.		
Lingual line:	convex with maximum convexity at the middle half.		
Anatomical landmarks:	*Contact area centered B-L below the crest of marginal ridge * <i>Sometimes the MMR is crossed by a marginal ridge groove.</i> *Deep developmental depression on the mesial surface of the root.	MMR is concave	
The Roots	Only the mesial root is seen because it is taller than the distal root		
Notes:	1- ML cusp is more pointed than MB cusp (because it is out of occlusion) 2- The cervical line slants occlusally from the buccal to the lingual side.		The buccal cusp tips are closer to the lingual cusp tips giving a small occlusal table

Distal Aspect	Lower 6	Lower 7	Lower 8
	The crown is shorter distally	* DBC is the shortest	
Convergence	There is a distal conversion , all 5 cusps could be seen.		
	*DMR is U- shape, shorter and more cervically than MMR *Distal contact area located below the distal cusp	*DMR is concave but less angular than MMR and more cervically located.	
N.B	Distal root is narrower than mesial root, so both roots are seen		

Occlusal Aspect	Lower 6	Lower 7	Lower 8
The geometric outline:	Hexagonal 	Rectangular 	Rectangular (similar to 7) or hexagonal (similar to 6) 
measurments	*larger MD than BL . * The crown converge lingually and distally		
Anatomical landmarks: Elevations:	*5 cusps with 5 triangular ridges *MBC is largest followed by the L cusps , DBC & DC. * MMR and DMR	*4 cusps with 4 triangular ridges (mesial cusps are larger than distal cusps). * Mesial and distal marginal ridge.	
Depressions:	*1 major central fossa + pit *2 Minor fossae (mesial and distal triangular fossae) **Central groove is zigzag in its MD coarse. **MB groove , DB and L groove	*1 major central fossa with central pit *2 Minor fossae (mesial and distal triangular fossae) *central groove is Cross pattern * supplemental grooves from developmental grooves	There are numerous supplemental grooves and pits giving a wrinkled appearance

Pulp cavity of lower 6

► Pulp Chamber :

- The pulp chamber is broad MD.
- It is almost rectangular in shape.
- There is a pulp horn beneath each cusp

► Root Canals :

- 3 main root canals.
- M root has 2 root canals while D root has 1 root canal.

Note: The two mesial root canals appear in x – ray overlapping each other.

Abnormalities you may find:

- 1- Three or four roots.
- 2- Four root canals.



Pulp cavity of lower 7 :

Pulp Chamber

- ► Bucco-Lingually, the pulp chamber is square
- ► Mesio-Distally, the pulp chamber is rectangular
- There is pulp horn beneath each cusp

Root Canals:

- 3 main root canals.
- M root usually has 2 root canals while D root has 1 root canal

- Note:

- **Cervical cross section:** Triangular due to small dimension distally
- **Mid root section:** *M root is kidney shape with 2 small canals
- *D root is oval with 1 large rounded canal



Note The pulp chamber of 8 is larger than other molars because it is younger than other molars by 9-11 years